Feature Dimensionality Reduction With $L_{2,p}$-Norm-Based Robust Embedding Regression for Classification of Hyperspectral Images

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***Abstract*— The curse of dimensionality and noise corruption are two tough problems that need to be solved in hyperspectral image (HSI) classification. However, the current feature dimensionality reduction methods, including both feature extraction (FE) and feature selection ones, cannot simultaneously solve the above two problems well. To address this issue, this article proposes a novel method called $L_{2,p}$-norm-based robust embedding regression ($L_{2,p}$-RER) for robust feature dimensionality reduction of HSI, which can effectively suppress the impact of noises and reduce the feature dimensions. Specifically, $L_{2,p}$-RER first integrates projection learning with robust principle component analysis (RPCA) to remove noise in a low-dimensional space. Second, an embedding regression (ER) regularization is proposed to improve the discriminability of the extracted low-dimensional features. Third, an $L_{2,1}$-norm constraint ($L_{2,1}$-NC) is imposed to improve the interpretability of the learned projection matrix, which can jointly extract the key features from all bands with their physical meanings certainly preserved. Last but most important, the $L_{2,p}$-norm that can adaptively balance the sparsity and the convexity is employed to model the noise and regression residual in the embedded low-dimensional space, which can further enhance the robustness and generalization of the proposed method. In addition, extensive experiments conducted on three benchmark HSI datasets validated the effectiveness of the proposed method.**

***Index Terms*— Embedding regression (ER), feature dimensionality reduction, hyperspectral image (HSI), $L_{2,p}$-norm.**

I. Introduction

With the development of hyperspectral remote sensing imaging technology, the use of hyperspectral images (HSIs) is becoming more common in the fields of environment monitoring [1], [2], [3], urban planning [4], [5], agriculture [6], [7], and so on. Its dense spectral sampling can acquire wealthy spectral information of objects, which is beneficial in distinguishing small differences of objects. In the past decades, two main types of classification methods, i.e., traditional machine learning [8], [9] and deep learning [10], [11], [12], have been developed, which have achieved well performance for HSI classification. Nevertheless, those methods fail to solve the curse of dimensionality, which is still a tough problem that needs to be further investigated in HSI classification. Feature dimensionality reduction has been a hot topic since it can effectively reduce information redundance and decrease the computational complexity for the classification of HSIs [13], [14]. Band selection (BS) (i.e., feature selection) and feature extraction (FE) are two major strategies for feature dimensionality reduction. BS reduces the dimensionality by selecting a subset from the original bands. Optimal clustering framework (OCF) and orthogonal-projection-based BS (OPBS) are representative methods [15], [16]. While FE attempts to derive a set of new bands or features from the original ones through some criterion functions, it can fully consider the information from all bands to avoid the loss of useful information.

Principal component analysis (PCA) [17] and linear discriminant analysis (LDA) [18] are two of the most popular FE methods. PCA aims at finding a projection direction to maximize the variance of the data. LDA seeks for an optimal discriminative subspace, in which the ratio of between-class distance to within-class distance is maximized. However, those linear FE methods fail to exploit the nonlinear structure of HSI data. To solve the problem, manifold leaning techniques that can reveal the nonlinear structure of data were proposed for the FE of HSIs. Isometric feature mapping (ISOMAP) [19], locally linear embedding (LLE) [20], and locality preserving projections (LPPs) [21] are typical manifold learning methods. LLE can automatically obtain a low-dimensional nonlinear manifold and then embed the data points into a low-dimensional embedding space. The main idea of LPP is to preserve the local geometric structure of HSI by using an adjacency graph. Nevertheless, these methods have an obvious drawback, that is, the L_2-norm would lead to an undesirable tendency of overemphasizing outliers in computing the projection matrix; thus, those L_2-norm-based methods are sensitive to noise.

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To solve the above issue, some robust criteria (i.e., L_1-norm, $L_{2,1}$-norm, L_p-norm, and $L_{2,p}$-norm) are used to replace L_2-norm for reducing the negative effect of noise and outliers. For example, L_1-PCA [22] and PCA-L_1 [23] are two representative ones of L_1-norm-based FE methods. The L_1-PCA minimizes the L_1-norm-based reconstruction error to acquire a projection matrix for FE, while PCA-L_1 maximizes the data variance with L_1-norm metric and solves the objective function by utilizing a greedy strategy. Motivated by those methods, LDA-L_1 [24] and LPP-L_1 [25] were proposed by employing L1-norm as metric. LDA-L_1 enhances the robustness of LDA by maximizing the L_1-norm-based between-class dispersion and minimizing the L_1-norm-based within-class dispersion simultaneously. LPP-L_1 employs the L_1-norm-based distance to measure the dissimilarity between pairs of points. In addition, the L_1-norm was generalized to L_p-norm for further enhancing the robustness of models [26]. For instance, Kwak [27] generalized PCA-L_1 to PCA-L_p, which maximizes the L_p-norm-based variance with an arbitrary value of $p > 0$. In this way, the conventional PCA and PCA-L_1 become two special cases of PCA-L_p with specific values of p. By selecting an appropriate p, the PCA-L_p method is able to obtain more satisfactory results. Similarly, LDA-L_p also utilizes the L_p-norm to enhance the performance of FE [28].

Since the $L_{2,1}$-norm can jointly utilize the intrinsic structure of data, it has also been a popular measure criterion for improving the robustness of FE methods. Up to now, numerous $L_{2,1}$-norm-based jointly sparse projection learning methods were proposed for FE of HSIs. The representative approaches include the minimum redundancy spectral feature (MRSF) selection [29], robust feature selection (RFS) [30], and unsupervised discriminative feature selection (UDFS) [31]. The MRSF is proposed to handle feature redundancy by adopting an embedded model with an $L_{2,1}$-norm constraint ($L_{2,1}$-NC). By imposing a joint $L_{2,1}$-norm on both the loss function and regularization term, Nie et al. [30] proposed the RFS for jointly sparse FE. UDFS [31] has also been proposed to achieve the jointly sparse projection via $L_{2,1}$-norm. Inspired by L_p-norm, Wang et al. [32] proposed $L_{2,p}$-PCA, which extends $L_{2,1}$-norm to $L_{2,p}$-norm, to minimize the reconstructed error. Due to the application of $L_{2,p}$-norm, $L_{2,p}$-PCA has the property of rotation invariance. However, the maximization of variance of data and the minimization of reconstruction error are equally significant. As a consequence, in order to jointly consider both the minimization of reconstruction error and the maximization of data variance in the feature space, the double $L_{2,p}$-norm-based PCA (DLPCA) was proposed [33].

As sparse and low-rank representation (LRR) can reveal the correlation among samples to eliminate redundancy, it has been widely applied for HSI analysis. For recovering the true low-dimensional subspace structure from noisy data, the robust PCA (RPCA) method was developed by decomposing the input data into a clean low-rank component and a sparse noise component [34]. Then, Liu et al. [35] proposed the LRR, whose main idea is to represent the high-dimensional data as the linear combination of a low-rank matrix. Li et al. [36] proposed an interpretable deep network by fully unfolding the LRR model, named LRR-Net. With the purpose of extending LRR for FE, low-rank embedding (LRE) was developed by integrating the LRR and projection learning into one model to deal with the corrupted image data [37]. To further reduce redundancy and improve robustness together, robust projection learning (RPL) was proposed to decompose HSI data into a low-rank component, a sparse part, and a noise term in a desired low-dimensional space [38]. Due to the excellent sparsity and robustness of sparse graph-embedding method, it is extensively employed on FE of HSIs. A typical example is the sparsity-preserving graph-embedding (SPGE) framework, which constructs an undirected weight graph according to the geometric characteristics of data [39]. Subsequently, to strengthen discriminant power, the sparse graph-based discriminant analysis (SGDA) was exploited by preserving the sparse connection in a block-structured affinity matrix with class-specific labeled samples [40]. Since low rankness and sparsity are equally significant in the construction of graph, the sparse and low-rank graph-based discriminant analysis (SLGDA) attempts to find an ideal projection matrix in an informative graph with both sparsity and low-rank constraints [41].

Without loss of generality, the prior information is important for FE as well as other tasks in image processing. Recently, linear regression regularization has been widely applied to capture the discriminative features to enhance the discriminability of FE methods. More importantly, those methods can well solve the small-class problem that has always bothered both RFS and LDA-based FE methods. Thus, some researchers propose to integrate linear regression and projection learning methods for constructing integrative FE models. Sparse principal component regression (SPCR) [42], joint SPCR (JSPCR) [43], and autoencoder regression (OAR) [44] are representatives. By taking both local geometrical structure and discriminative information into consideration, Lai et al. [45] integrated the properties of LPP and ridge regression and proposed generalized robust regression (GRR). To eliminate the limitation of the number of projections and learn a discriminative subspace, Lai et al. [46] proposed discriminative low-rank projection (DLRP) by combining the properties of LRR and least-squares regression (LSR). Similarly, the locality preserving robust regression (LPRR) method [47] was developed by integrating the advantages of locality preservation and linear regression based on capped-L_2 norm into a unified framework. Low-rank discriminative regression learning (LDRL) [48] was proposed to enhance the robustness of FE via making full use of the underlying low rankness and sparsity of data. The generalized embedding regression (GER) framework [49] was constructed to simultaneously achieve low-dimensional embedding and sparse projection learning in a joint objective function with a generalized orthogonal constraint. The low-rank preserving embedding regression (LRPER) [50] method integrated LRR, linear regression, and projection learning into a unified framework to improve the discriminability and robustness of projection learning.

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However, the previous works mainly focus on utilizing linear regression to improve the discriminability of FE methods, whose robustness and adaptive ability are still unsatisfactory for HSI analysis. To address the above issue, this article proposes a robust feature dimensionality reduction method called $L_{2,p}$-norm-based robust embedding regression ($L_{2,p}$-RER) for HSI classification, which can adaptively and effectively suppress noise and reduce the feature dimensions. In brief, the main advantages of $L_{2,p}$-RER can be summarized as follows.

1) The proposed $L_{2,p}$-RER method integrates projection learning with RPCA to recover the clean data and remove noise in a low-dimensional space, which ensures the robustness of learned projection.
2) An embedding regression (ER) term is used to search the relationship between the mapped low-dimensional features and labels in a training set to improve the discriminability of projection learning.
3) $L_{2,p}$-norm is introduced to model the noise term and regression residual in the embedded low-dimensional space, which can adaptively balance the sparsity and the convexity to enhance the robustness and generalization of the proposed method.
4) $L_{2,1}$-NC is employed to learn a row sparse projection matrix, which makes $L_{2,p}$-RER capable of FE and feature selection simultaneously. Moreover, the learned projection can jointly select important features from the original ones to compose the low-dimensional subspace, which is more interpretable.

II. Related Work

A. Notations

In this article, $X = [x_1, x_2, \ldots, x_n] \in R^{d\times n}$ denotes the observation data and $Y = [y_1, y_2, \ldots, y_n] \in R^{c\times n}$ denotes the label matrix, where c is the total number of classes in the training samples. $Y_{ij} = 1$ if x_i belongs to the jth class and 0 otherwise. The $L_{2,p}$-norm of the matrix X is defined as

$$\|X\|_{2,p} = \left(\sum_{i=1}^{d}\left(\sum_{j=1}^{m}|x_{i,j}|^2\right)^{\frac{p}{2}}\right)^{\frac{1}{p}} = \left(\sum_{i=1}^{d}\left\|x^i\right\|_2^p\right)^{\frac{1}{p}}. \quad (1)$$

B. Robust PCA

The PCA assumes that the noise of data is Gaussian, and its effectiveness is influenced by heavy noise or outliers. To improve the robustness of PCA, RPCA was proposed, whose goal is to decompose data X to $L + E$, where L is the low-rank component and E denotes the sparse error. Its objective function is given as follows:

$$\min_{L,\ E}\ \operatorname{rank}(L) + \lambda\|E\|_0 \quad \text{s.t. } X = L + E \quad (2)$$

where $\operatorname{rank}(L)$ means the rank of L and λ is a positive parameter to trade off the low rankness and sparsity.

C. Robust Feature Selection

Since the L_2-norm-based methods are sensitive to outliers and L_1-norm-based methods cannot obtain the jointly sparse projection. Nie et al. [30] proposed RFS via joint $L_{2,1}$-norms minimization. By utilizing the $L_{2,1}$-norm on both loss function and regularization term, RFS can release the drawback in ridge regression for its sensitivity to outliers

$$\min_{B}\ \|Y - \mathrm{XB}\|_{2,1} + \beta\|B\|_{2,1} \quad (3)$$

where β is a balance parameter that needs to be turned, Y is the label matrix, and B denotes the regression matrix.

III. Methodology

A. $L_{2,p}$-Norm-Based Robust ER

In practice, HSIs are inevitably contaminated by noise during their acquisition, which further complicates HSI data processing. It is well known that RPCA is an effective method to eliminate the influence of noise because of its robustness. However, RPCA mainly focuses on decreasing noise interference without the ability to reduce dimensionality. To make full use of its robustness, we attempt to construct a model that can maintain the robustness of RPCA for robust feature dimensionality reduction. Thus, based on the assumption that data can also be decomposed into a low-rank component and sparse noise part in a low-dimensional subspace, the projection learning is integrated into the RPCA model to obtain a more robust projection matrix. Mathematically, the model can be formulated as

$$\min_{P,\ L}\ \operatorname{rank}(L) + \|P^TX - P^TL\|_0 \quad \text{s.t. } P^TP = I \quad (4)$$

where $P \in R^{d\times m}$ is a projection matrix, $P^TX - P^TL$ represents a sparse structural noise, and L denotes a low-rank matrix.

For the FE of HSIs, the prior label information plays a critical role in improving the discriminability of features. As mentioned before, linear regression regularization has been widely used to improve the discriminability of FE methods. However, they often suffer from the so-called small-class problem, while linear regression is applied for FE directly. In order to address the limitation of the maximum number of projections, an ER term is formulated to exploit the mapping relationship between the embedded low-dimensional feature space and the label space rather than that between the raw data space and the label space. Then, the objective function after adding the regression term is

$$\min_{P,\ L,\ W}\ \operatorname{rank}(L) + \left\|P^TX - P^TL\right\|_0 + \left\|Y - WP^TX\right\|_F^2 \quad \text{s.t. } P^TP = I,\quad W^TW = I \quad (5)$$

where $W \in R^{c\times m}$ is the orthogonal matrix.

Similar to the existing projection learning methods, the projection learned by (5) assumes that all bands have the same importance, which fails to consider their different importance. Moreover, the extracted low-dimensional features lose the

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physical meaning of the original bands. Therefore, the $L_{2,1}$-NC is imposed on the projection matrix to obtain the key features from all of the original ones. In this way, the learned projection matrix is more interpretable. Then, we have

$$\begin{aligned}\min_{P,\ L,\ W}\ & \operatorname{rank}(L)+\left\|P^TX-P^TL\right\|_0\\ &+\left\|Y-WP^TX\right\|_F^2+\|P\|_{2,1}\\ \text{s.t. }& P^TP=I,W^TW=I.\end{aligned}\tag{6}$$

To further enhance the robustness and adaptive ability for the FE of HSIs, $L_{2,p}$-norm is employed to model the noise term and regression residual, which can adaptively balance sparsity and convexity. Specifically, the value of p can be adaptively selected for different HSI datasets, which can improve the universality of the model. Moreover, to balance the influence between various terms, three regularization parameters are introduced. Consequently, the exact formulation of the proposed $L_{2,p}$-RER model is

$$\begin{aligned}\min_{P,\ L,\ W}\ & \operatorname{rank}(L)+\alpha\left\|P^TX-P^TL\right\|_{2,p}^p\\ &+\beta\left\|Y-WP^TX\right\|_{2,p}^p+\gamma\|P\|_{2,1}\\ \text{s.t. }& P^TP=I,\quad W^TW=I\end{aligned}\tag{7}$$

where α, β, and γ are parameters to balance different terms.

Since the rank function minimizing is NP-hard, inspired by [34], the problem can be effectively solved by relaxing the rank function with its convex lower bound $\|\cdot\|_*$. Thus, the problem (7) turns into the nuclear norm minimization problem

$$\begin{aligned}\min_{P,\ L,\ W}\ & \|L\|_*+\alpha\left\|P^TX-P^TL\right\|_{2,p}^p\\ &+\beta\left\|Y-WP^TX\right\|_{2,p}^p+\gamma\|P\|_{2,1}\\ \text{s.t. }& P^TP=I,\quad W^TW=I.\end{aligned}\tag{8}$$

B. Optimization

An alternating iteration algorithm is proposed to obtain the optimal solution of problem (8). From the definition of $L_{2,p}$-norm, problem (8) is equivalent to

$$\begin{aligned}\min_{P,\ L,\ W}\ & \|L\|_*+\alpha\sum_{i=1}^n\left\|P^Tx_i-P^Tl_i\right\|_2^p\\ &+\beta\sum_{i=1}^n\left\|y_i-WP^Tx_i\right\|_2^p+\gamma\|P\|_{2,1}\\ \text{s.t. }& P^TP=I,\quad W^TW=I.\end{aligned}\tag{9}$$

If $b_i=(p/2)\left\|P^Tx_i-P^Tl_i\right\|_2^{p-2}$ and $f_i=(p/2)\left\|y_i-WP^Tx_i\right\|_2^{p-2}$, then problem (9) can be reformulated into

$$\begin{aligned}\min_{P,\ L,\ W}\ & \|L\|_*+\alpha\sum_{i=1}^n\|P^Tx_i-P^Tl_i\|_2^2b_i\\ &+\beta\sum_{i=1}^n\|y_i-WP^Tx_i\|_2^2f_i+\gamma\|P\|_{2,1}\\ \text{s.t. }& P^TP=I,\quad W^TW=I.\end{aligned}\tag{10}$$

Equivalently, the above problem can be written in the following matrix form:

$$\begin{aligned}\min_{P,\ L,\ W}\ & \|L\|_*+\alpha\left\|P^TX\sqrt{B}-P^TL\sqrt{B}\right\|_F^2\\ &+\beta\left\|Y\sqrt{F}-WP^TX\sqrt{F}\right\|_F^2+\gamma\|P\|_{2,1}\\ \text{s.t. }& P^TP=I,\quad W^TW=I\end{aligned}\tag{11}$$

where $B=\operatorname{diag}(b_1,b_2,\ldots,b_n)$ and $F=\operatorname{diag}(f_1,f_2,\ldots,f_n)$ are diagonal matrices.

The optimization problem (11) can be solved by the augmented Lagrangian method (ALM) algorithm. The augmented Lagrangian function of (11) is

$$\begin{aligned}\mathcal{L}=&\|M\|_*+\alpha\|E\|_F^2+\beta\|Q\|_F^2\\ &+\frac{\mu}{2}\left(\left\|L-M+\frac{\omega_1}{\mu}\right\|_F^2\right.\\ &+\left\|E-P^TX\sqrt{B}+P^TL\sqrt{B}+\frac{\omega_2}{\mu}\right\|_F^2\\ &\left.+\left\|Q-Y\sqrt{F}+WP^TX\sqrt{F}+\frac{\omega_3}{\mu}\right\|_F^2\right)+\gamma\|P\|_{2,1}\end{aligned}\tag{12}$$

where $M=L$, $E=P^TX(B)^{1/2}-P^TL(B)^{1/2}$, $Q=Y(F)^{1/2}-WP^TX(F)^{1/2}$, and ω_1, ω_2, and ω_3 are the Lagrangian multipliers.

Thus, the optimal M can be updated by

$$\min_M\ \|M\|_*+\frac{\mu}{2}\left\|L-\left(M+\frac{\omega_1}{\mu}\right)\right\|_F^2\tag{13}$$

which can be solved by the singular value thresholding (SVT) operator as

$$M=\Theta_{\frac{1}{\mu}}\left(L+\frac{\omega_1}{\mu}\right).\tag{14}$$

Updating L by minimizing (12) is equivalent to the following problem:

$$\begin{aligned}\min_L\ & \left\|L-M+\frac{\omega_1}{\mu}\right\|_F^2\\ &+\left\|E-P^TX\sqrt{B}+P^TL\sqrt{B}+\frac{\omega_2}{\mu}\right\|_F^2.\end{aligned}\tag{15}$$

Taking the derivative with respect to L and setting it to zero, the following formulation can be obtained:

$$\begin{aligned}&2L-N_1+LB+PN_2\sqrt{B}=0\\ &\Rightarrow L=\left(N_1-PN_2\sqrt{B}\right)(2I+B)^{-1}\end{aligned}\tag{16}$$

where $N_1=M+(\omega_1/\mu)$ and $N_2=E-P^TX(B)^{1/2}+(\omega_2/\mu)$. In this way, an optimal L can be estimated.

Then, by fixing other variables than E, problem (12) can be converted into the following equivalent problem:

$$\min_E\ \alpha\|E\|_F^2+\frac{\mu}{2}\left\|E-P^TX\sqrt{B}+P^TL\sqrt{B}+\frac{\omega_2}{\mu}\right\|_F^2.\tag{17}$$

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Taking the partial derivative of E, the update equation becomes

$$2\alpha E + \mu E - \mu\left(P^T X\sqrt{B} - P^T L\sqrt{B} - \frac{\omega_2}{\mu}\right) = 0$$
$$\Rightarrow E = \frac{\mu}{2\alpha+\mu}\left(P^T X\sqrt{B} - P^T L\sqrt{B} - \frac{\omega_2}{\mu}\right). \quad (18)$$

Fixed other variables, Q can be solved by minimizing the following problem:

$$\min_{Q} \ \beta\|Q\|_F^2 + \frac{\mu}{2}\left\|Q - Y\sqrt{F} + WP^T X\sqrt{F} + \frac{\omega_3}{\mu}\right\|_F^2 \quad (19)$$

and it can be solved similar to the problem (15), namely

$$2\beta Q + \mu Q - \mu\left(Y\sqrt{F} - WP^T X\sqrt{F} - \frac{\omega_3}{\mu}\right) = 0$$
$$\Rightarrow Q = \frac{\mu}{2\beta+\mu}\left(Y\sqrt{F} - WP^T X\sqrt{F} - \frac{\omega_3}{\mu}\right). \quad (20)$$

According to the definition of the $L_{2,1}$-norm, if we fix other variables except P, the problem (12) can be rewritten as

$$\min_{P} \ \frac{\mu}{2}\left(\left\|E - P^T X\sqrt{B} + P^T L\sqrt{B} + \frac{\omega_2}{\mu}\right\|_F^2 + \left\|Q - Y\sqrt{F} + WP^T X\sqrt{F} + \frac{\omega_3}{\mu}\right\|_F^2\right) + \gamma tr\left(P^T DP\right) \quad (21)$$

where $D \in R^{d\times d}$ is a diagonal matrix with the ith element being $D_{ii} = (1/2\|p^i\|_2)$. Then, computing the gradient of (21) with respect to P and setting it as zero

$$\left(C_1 C_1^T + \mathrm{XFX}^T + \frac{2\gamma}{\mu}D\right)P - C_2 C_1^T + W^T C_3\sqrt{F}X^T = 0 \quad (22)$$

where $C_1 = X(B)^{1/2} - L(B)^{1/2}$, $C_2 = E + (\omega_2/\mu)$, and $C_3 = Q - Y(F)^{1/2} + (\omega_3/\mu)$. Thus, the optimal P is

$$P = \left(C_2 C_1^T - W^T C_3\sqrt{F}X^T\right) * \left(C_1 C_1^T + \mathrm{XFX}^T + \frac{2\gamma}{\mu}D\right)^{-1}. \quad (23)$$

By fixing other variables than W, problem (11) becomes

$$\min_{W} \ \left\|Y\sqrt{F} - WP^T X\sqrt{F}\right\|_F^2 \quad (24)$$

which is equivalent to

$$\max_{W} \ tr\left(WP^T \mathrm{XFY}^T\right). \quad (25)$$

Suppose that the SVD of $P^T\mathrm{XFY}^T$ is USV^T. The solution to (25) is $W = UV^T$. The iterative algorithm for solving L_2, p-RER is summarized in Algorithm 1.

Algorithm 1 Solving Algorithm for $L_{2,p}$-RER

Input: Data $X \in R^{d\times n}$, label matrix $Y \in R^{c\times n}$, reduced dimension $m(\leq d)$, parameter α, β, and γ.
Initialization: Initialize $L = E = Q = 0$ and $\omega_1 = \omega_2 = \omega_3 = \mathbf{0}$, where $\mathbf{0}$ is zero matrix. Initialize $D \in R^{d\times d}$, $B \in R^{n\times n}$, $P \in R^{d\times m}$, as identity matrix, respectively.
while not converged **do**
Step1: Update M by computing (14).
Step2: Update L by computing (16).
Step3: Update E by computing (18).
Step4: Update Q by computing (20).
Step5: Update P by computing (23).
Step6: Update W by optimizing (25).
Step7: Update D, B, F.
end while
Output: The projection matrix P

TABLE I
CLASS LABELS AND TRAIN-TEST DISTRIBUTION OF SAMPLES FOR THE UNIVERSITY OF PAVIA DATASET

#	Class	Train	Test
1	Asphalt	66	6565
2	Meadows	186	18463
3	Gravel	20	2079
4	Trees	30	3034
5	Painted Metal Sheets	13	1332
6	Bare Soil	50	4979
7	Bitumen	13	1317
8	Self-Blocking Bricks	36	3646
9	Shadows	10	937
	Total	424	42352

IV. EXPERIMENTS

This section first introduces three HSI datasets involved in the experiments. Then, the impact of parameters in the proposed $L_{2,p}$-RER method is analyzed. To study the distribution of extracted features, the t-distributed stochastic neighbor embedding (t-SNE) is utilized to present visual experimental results of FE yielded by the $L_{2,p}$-RER and comparative methods. Meanwhile, the OAs along with different dimensions are calculated to explore an optimal number of dimension for subsequent experiments. Finally, some comparative experiments are conducted and the corresponding results analyses are presented.

A. Data Description

1) University of Pavia Dataset: The HSI dataset was obtained by the German airborne reflection optical spectral imager. It contains 610 × 340 pixels with 103 spectral bands covering the spectral range from 0.43 to 0.86 μm and nine-land-cover categories. Only 1% samples are utilized to train for the University of Pavia dataset. The detailed quantity of training and testing samples is summarized in Table I.

2) Salinas Dataset: The Salinas dataset was collected by the Airborne Visible Infrared Imaging Sensor (AVIRIS) over Salinas Valley, Southern CA, in 1998, with a spatial resolution of 3.7 m. The image comprises 512 × 217 pixels with 204 bands after 20 water-absorption bands are removed. It has

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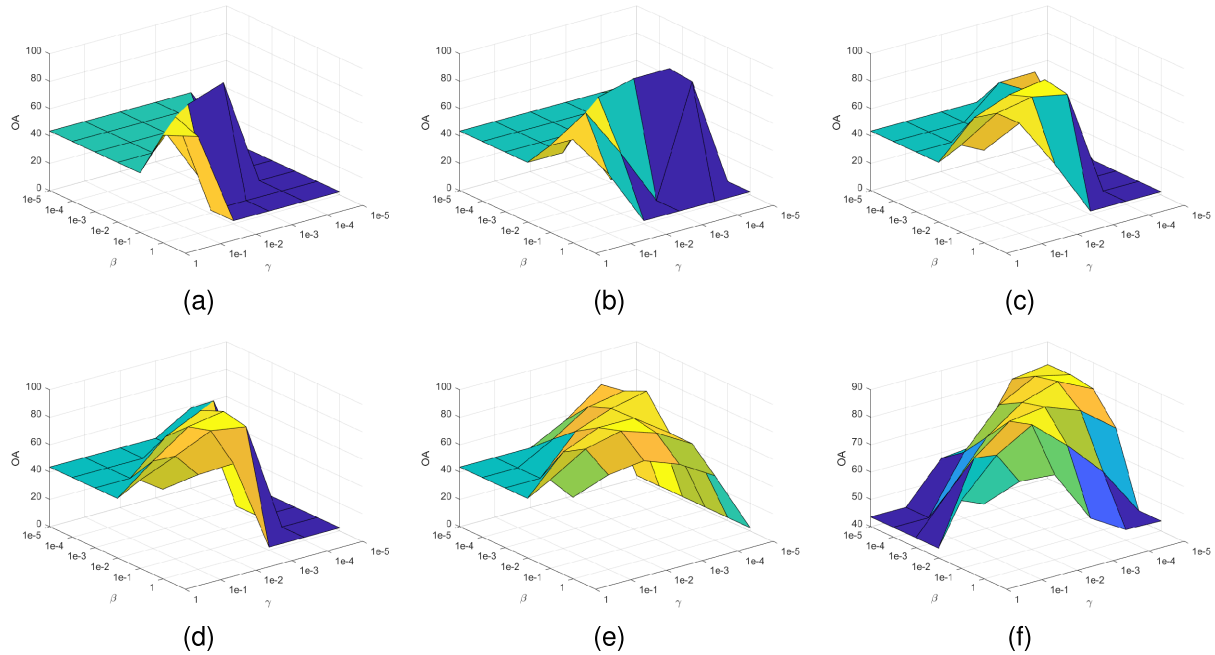


Fig. 1. Variation of OA with β and γ when α is set as different values for the University of Pavia dataset. (a) $\alpha = 1$. (b) $\alpha = 0.1$. (c) $\alpha = 0.01$. (d) $\alpha = 0.001$. (e) $\alpha = 0.0001$. (f) $\alpha = 0.00001$.

TABLE II
CLASS LABELS AND TRAIN–TEST DISTRIBUTION OF SAMPLES FOR THE SALINAS DATASET

#	Class	Train	Test
1	Brocoli-green-weeds-l	20	1989
2	Brocoli-green-weeds-2	37	3689
3	Fallow	19	1957
4	Fallow-rough-plow	13	1381
5	Fallow-smooth	26	2652
6	Stubble	39	3920
7	Celery	35	3544
8	Grapes-untrained	112	11159
9	Soil-vinyard-develop	62	6141
10	Corn-senesced-green-weeds	32	3246
11	Lettuce-romaine-4wk	10	1058
12	Lettuce-romaine-5wk	19	1908
13	Lettuce-romaine-6wk	10	906
14	Lettuce-romaine-7wk	10	1060
15	Vinyard-untrained	72	7196
16	Vinyard-vertical-trellis	18	1789
	Total	534	53595

16 classes, which mainly contains vegetables, bare soils, and vineyard fields. There are 1% labeled samples for training and the rest for testing. The numbers of training and testing samples are listed in Table II.

3) Heihe Dataset: The imagery was gathered by the CASI/SASI sensors in the Zhangye basin, which is located in the middle reaches of Heihe watershed, Gansu Province, China. The size of the dataset is 684 × 453 and with eight types of land covers. After removing 14 bands that are severely affected by noise, 135 available bands were introduced in the experiments. There are 20 samples of each land cover as the training samples for the Heihe dataset. More detailed information on the number of training and testing samples is displayed in Table III.

TABLE III
CLASS LABELS AND TRAIN–TEST DISTRIBUTION OF SAMPLES FOR THE HEIHE DATASET

#	Class	Train	Test
1	Corn	20	41051
2	Fragrant-flowered Garlic	20	28566
3	Cauliflower	20	20335
4	Bell Pepper	20	7588
5	Potato	20	3742
6	Endive Sprout	20	1655
7	Watermelon	20	965
8	Artificial Surface	20	855
	Total	160	104757

B. Experimental Setup

In our experiments, to validate the performance of the proposed method, several related FE methods, such as LPP, SLGDA, LRE, LPRR, DLRP, and RPL, are employed for comparison. To quantitatively evaluate $L_{2,p}$-RER, per-class accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) are used as evaluation criteria. Since this article focuses on FE, the classical support vector machine (SVM) classifier is adopted to conduct the classification. In particular, to avoid bias, each method is rerun 50 times and the average values as well as the standard deviation are reported.

For the $L_{2,p}$-RER model, there are four underlying parameters (i.e., α, β, γ, and p) that need to be predetermined. A cross-validation strategy is employed for selecting the suitable parameters. The reasonable values of the parameters α, β, and γ are chosen from the set of $\{0.00001, 0.0001, 0.001, 0.01, 0.1, 1\}$. The acceptable value of p is selected from the range of $\{0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2\}$. Specifically,

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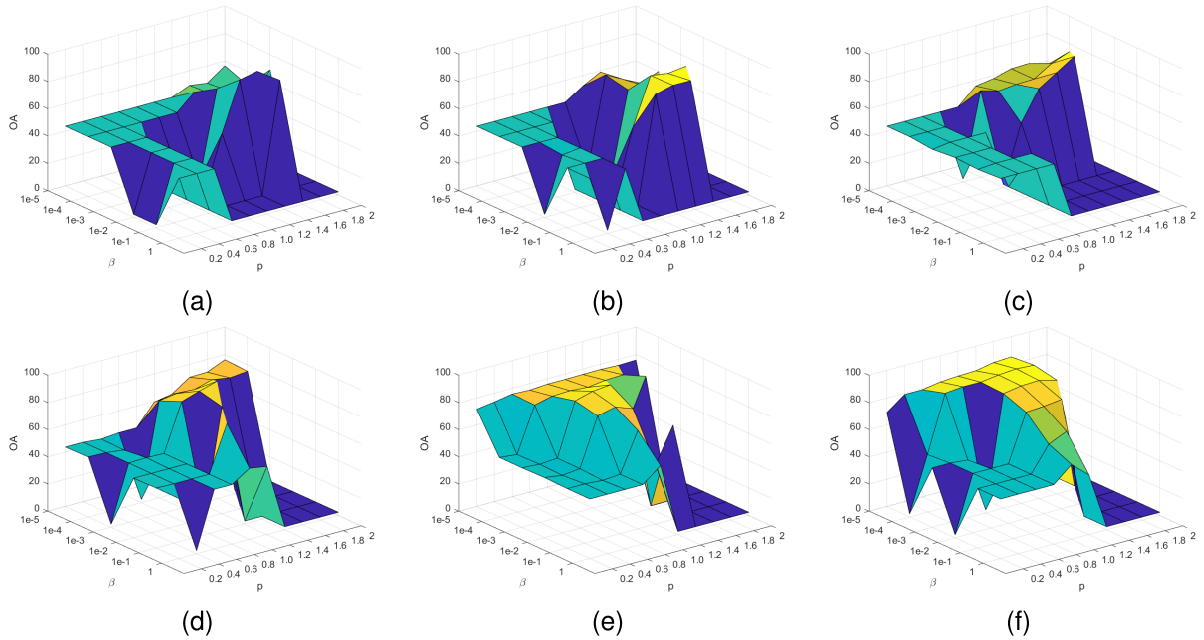


Fig. 2. Variation of OA with β and p when α is set as different values for the University of Pavia dataset. (a) $\alpha = 1$. (b) $\alpha = 0.1$. (c) $\alpha = 0.01$. (d) $\alpha = 0.001$. (e) $\alpha = 0.0001$. (f) $\alpha = 0.00001$.

we take the parameter tuning for the University of Pavia dataset as an example. Fig. 1 shows the variation curves of OA with β and γ when α is set as different values and the changes of OA with β and p under different values of α are shown in Fig. 2. From those figures, it can be seen that setting α and β as 0.0001 and 0.001, respectively, is appropriate. Furthermore, we can draw the conclusion that the reasonable values for γ and p can be set as 0.0001 and 1.2 according to Figs. 1(e) and 2(e), respectively. Thus, the parameters of $L_{2,p}$-RER are set as $\{\alpha,\ \beta,\ \gamma,\ p\} = \{0.0001,\ 0.001,\ 0.0001,\ 1.2\}$ for the University of Pavia dataset. In the same way, the reasonable values of α, β, γ, and p for the Salinas and Heihe datasets can also be found. In detail, the acceptable values of $\{\alpha,\ \beta,\ \gamma,\ p\}$ of $L_{2,p}$-RER are set as $\{0.1,\ 1,\ 0.1,\ 0.8\}$ for Salinas dataset. For the Heihe dataset, setting $\{\alpha,\ \beta,\ \gamma,\ p\} = \{0.1,\ 0.01,\ 0.001,\ 1.4\}$ is reasonable. Moreover, the parameters of the comparative methods are also adjusted to the optimum according to the associated references.

C. Feature Distribution Visualization

To visually compare the effectiveness of the proposed method, the features projected by different methods are visualized by using t-SNE. The distributions of the low-dimensional features for three HSI datasets in 2-D space are shown in Figs. 3–5. Samples from the same category are visualized by the same color, and samples from different categories are represented by different colors. From the visualization results, it can be observed whether each method sufficiently distinguishes each category. Obviously, the distribution maps obtained by LPP, SLGDA, LRE, and RPL contain many overlapping areas, which means that these methods are not able to discriminate samples from different categories. The main reason for this result may be that those unsupervised methods do not take into account the prior information of the HSI datasets. It can be seen that LPRR and DLRP perform better than those unsupervised methods, while there still are some small overlapping areas in their distribution maps. The proposed $L_{2,p}$-RER produces the best intraclass aggregation and interclass separability in the distribution maps among all of those methods, which enhances class separability.

D. Sensitivity Analysis of Feature Dimensions

To detailedly explore the influence of feature dimensionality reduction on classification accuracy, the numerical simulation experiments for the sensitivity analysis of feature dimensions are conducted on the University of Pavia, Salinas, and Heihe datasets by using $L_{2,p}$-RER and the comparative FE methods. Considering that the purpose of FE is to extract low-dimensional representative features from the original data, the feature dimensions from the range of [1, 30] are studied in the experiments. The corresponding experimental results of three HSI datasets are presented in Fig. 6.

In particular, Fig. 6(a) shows the changes of OA along with the feature dimensions for the University of Pavia dataset. It can be seen that the accuracy of all methods increases sharply with the dimensionality increasing when the dimension is less than 10. While the dimension is larger than 10, the increasing of OA slows down and tends to stable. Obviously, the proposed $L_{2,p}$-RER method achieves the highest OA, while the dimension of features is larger than 5.

Fig. 6(b) and (c) show the relationships between feature dimensions and classification accuracy for the Salinas and Heihe datasets, respectively. From Fig. 6(b) and (c), a similar conclusion can be drawn that the OA tends to stable with the feature dimension increasing for all involved FE methods. More importantly, $L_{2,p}$-RER also yields better classification performance than the comparative methods when the OA is stable. In addition, for the three datasets, setting the final

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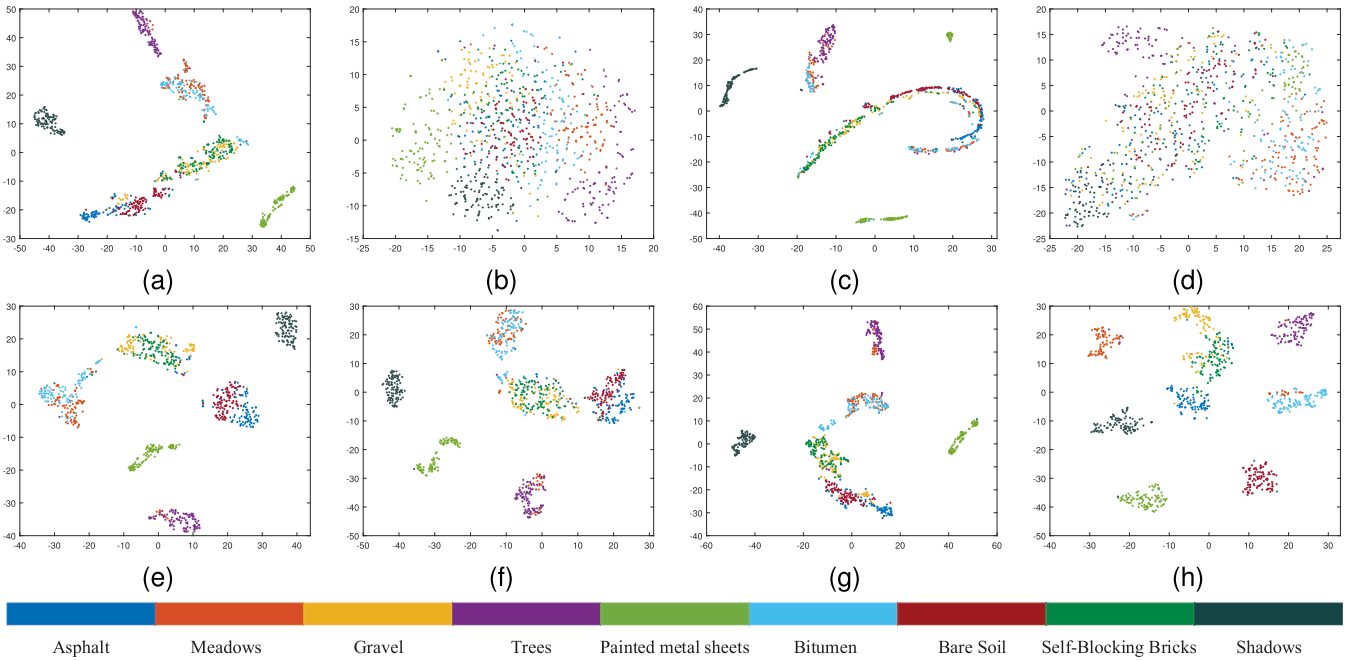


Fig. 3. Results of projection into 2-D space by different methods on the University of Pavia dataset. (a) Original. (b) LPP. (c) SLGDA. (d) LRE. (e) LPRR. (f) DLRP. (g) RPL. (h) $L_{2,p}$-RER ($p = 1.2$).

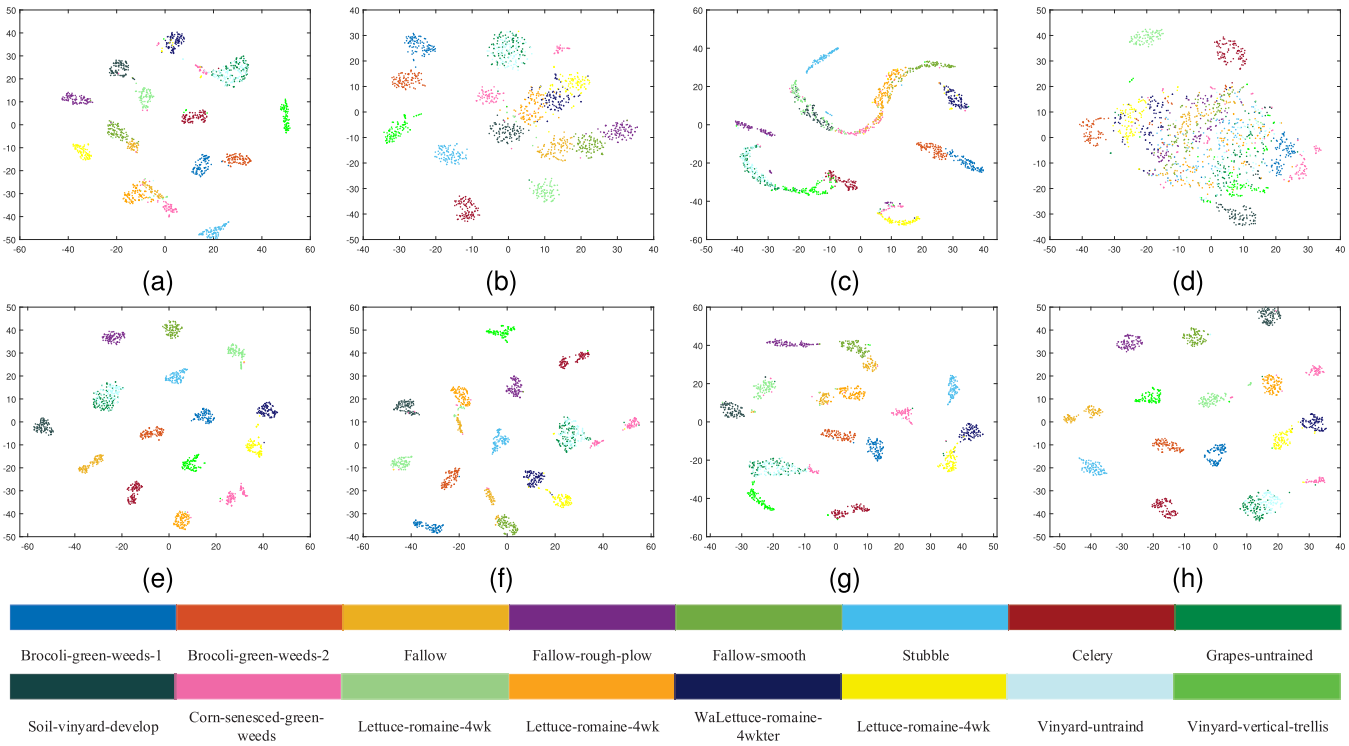


Fig. 4. Results of projection into 2-D space by different methods on the Salinas dataset. (a) Original. (b) LPP. (c) SLGDA. (d) LRE. (e) LPRR. (f) DLRP. (g) RPL. (h) $L_{2,p}$-RER ($p = 0.8$).

extracted feature dimension as 30 is reasonable in the following experiments.

E. Comparison of Classification Performance

The quantitative evaluation metrics, such as per-class accuracy, OA, AA, and κ, obtained by different FE methods for the involved datasets are presented in Tables IV–VI and the associated classification maps are displayed in Figs. 7–9. On the whole, the performance of $L_{2,p}$-RER is better than the comparative methods for all datasets. The experimental results of the University of Pavia dataset are shown in Table IV. It is obvious that the proposed $L_{2,p}$-RER improves the accuracy of OA by 6.26%, 6.25%, 4.80%, 3.84%, 3.51%, and 1.75%

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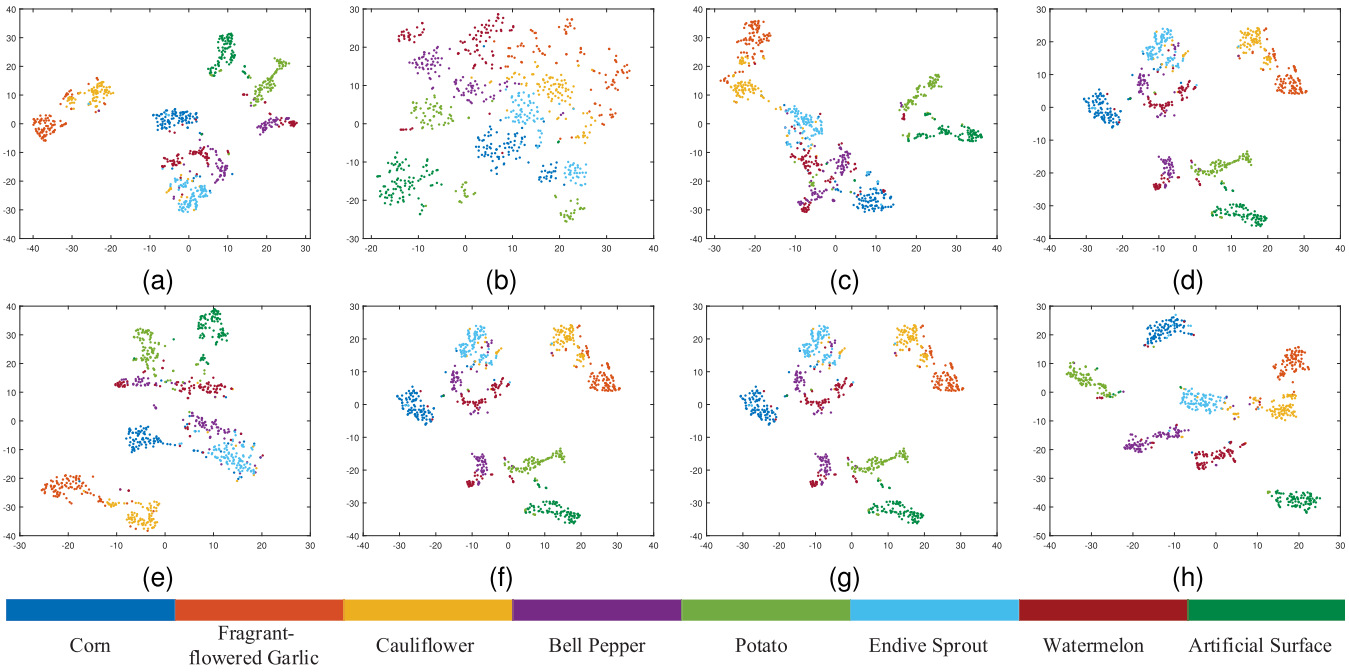


Fig. 5. Results of projection into 2-D space by different methods on the Heihe dataset. (a) Original. (b) LPP. (c) SLGDA. (d) LRE. (e) LPRR. (f) DLRP. (g) RPL. (h) $L_{2,p}$-RER ($p = 1.4$).

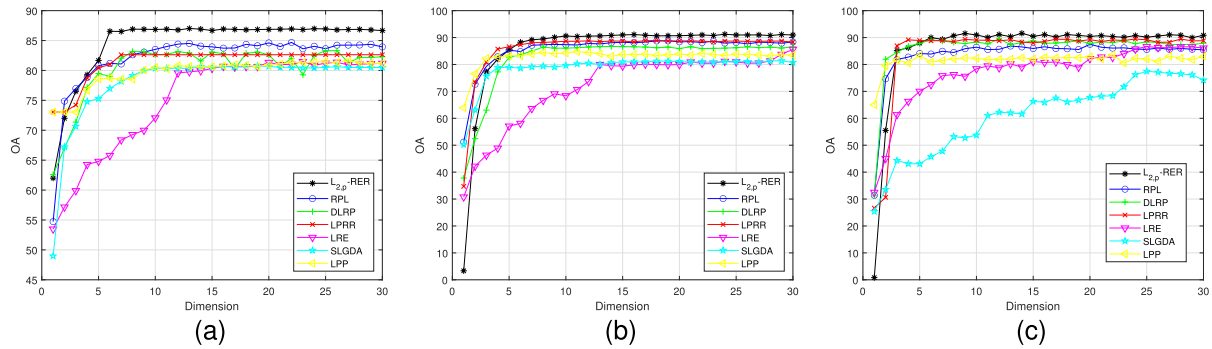


Fig. 6. OA versus variations of dimension for different methods. (a) University of Pavia dataset. (b) Salinas dataset. (c) Heihe dataset.

TABLE IV
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR THE UNIVERSITY OF PAVIA DATASET USING THE SVM CLASSIFIER

Class	LPP	SLGDA	LRE	LPRR	DLRP	RPL	$L_{2,p}$-RER (p=1.2)
1	84.16±2.56	**90.72±3.22**	80.79±2.13	80.41±4.18	85.95±2.52	85.20±1.93	88.06±2.30
2	94.43±1.32	93.03±1.31	93.89±1.16	94.22±4.72	92.03±1.79	94.11±1.21	**94.91±1.66**
3	56.50±6.93	50.77±6.89	52.85±5.46	57.12±3.98	55.21±6.57	**62.52±7.48**	61.02±6.57
4	59.55±5.38	**85.63±4.92**	84.13±4.05	75.73±4.03	78.51±4.00	77.86±4.78	80.82±3.60
5	50.98±7.41	93.46±5.32	98.39±0.79	97.07±0.19	95.41±3.95	77.57±8.37	**98.44±1.27**
6	53.94±5.31	58.18±5.06	56.11±4.27	57.86±4.68	63.96±6.22	**68.31±3.54**	67.27±5.31
7	72.34±6.86	27.52±7.68	55.31±8.83	58.09±4.72	64.98±8.29	**74.97±6.45**	71.31±8.85
8	78.53±4.01	62.81±6.00	74.45±5.14	82.06±4.99	75.20±4.48	79.78±4.27	**84.01±4.77**
9	91.05±5.10	42.00±7.12	80.44±8.37	96.38±0.24	98.18±1.59	98.19±1.43	**99.44±0.35**
OA	80.21±1.24	80.22±1.37	81.67±0.75	82.63±2.18	82.96±1.47	84.72±0.87	**86.47±0.79**
AA	71.27±2.07	67.12±5.70	75.15±1.47	77.66±1.09	78.83±1.94	79.83±1.53	**82.81±1.42**
κ	72.64±1.81	73.19±1.96	75.35±1.01	76.61±2.60	77.17±2.00	79.44±1.21	**81.83±1.07**

compared to LPP, SLGDA, LRE, LPRR, DLRP, and RPL, respectively, In addition, accuracies of AA and κ also have a certain improvement. From Fig. 7, a similar conclusion can be obtained for the University of Pavia dataset, i.e., there are fewer misclassification areas on the classification map of the $L_{2,p}$-RER. From Table V, it can be seen that the proposed $L_{2,p}$-RER also yields the best OA, AA, and κ on the Salinas dataset. The gain of $L_{2,p}$-RER over those unsupervised FE methods (such as LPP, SLGDA, LRE, and RPL) is at least 2%, which implies that the prior information is significant for improving the discriminative ability of low-dimensional features. In particular, $L_{2,p}$-RER almost achieves the highest

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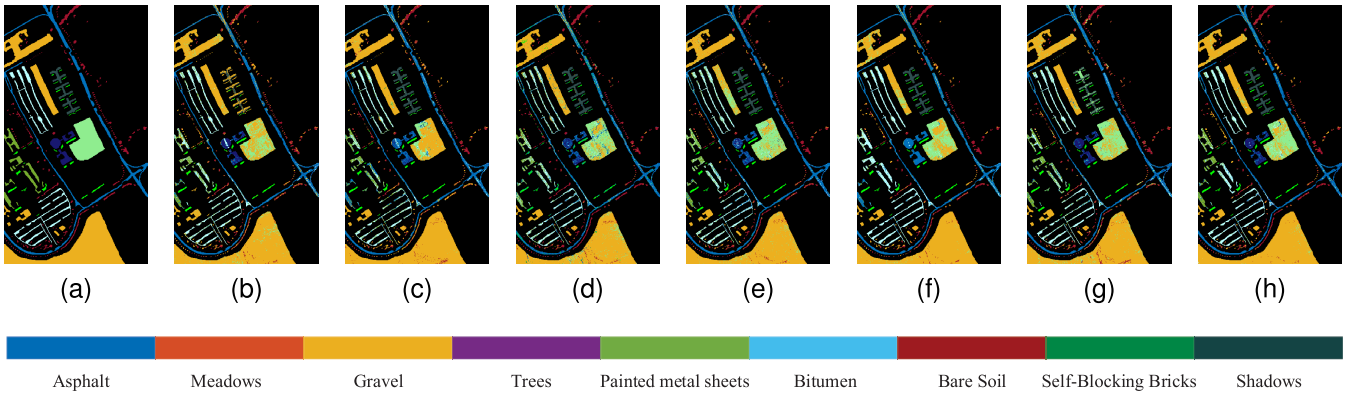


Fig. 7. Classification maps of different methods for the University of Pavia dataset. (a) Original. (b) LPP. (c) SLGDA. (d) LRE. (e) LPRR. (f) DLRP. (g) RPL. (h) $L_{2,p}$-RER ($p = 1.2$).

TABLE V
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR THE SALINAS DATASET USING THE SVM CLASSIFIER

Class	LPP	SLGDA	LRE	LPRR	DLRP	RPL	$L_{2,p}$-RER (p=0.8)
1	95.52±1.81	97.02±1.51	98.62±0.94	99.16±0.74	97.47±1.21	97.75±0.82	**99.29±0.84**
2	93.74±2.15	99.23±0.27	98.76±1.42	99.18±0.53	99.63±0.25	99.25±0.48	**99.84±0.15**
3	89.48±5.76	67.80±15.16	88.84±6.03	98.54±1.90	81.88±3.65	94.53±2.78	**99.31±0.65**
4	87.27±6.18	94.50±5.38	96.29±2.92	98.80±2.02	97.46±1.51	98.08±2.45	**99.13±0.71**
5	95.42±1.12	93.77±2.29	96.71±1.33	97.66±0.75	93.86±2.19	97.72±0.87	**98.52±0.60**
6	93.16±2.88	97.83±1.91	89.77±3.90	99.34±0.62	99.67±0.16	99.34±0.40	**99.76±0.11**
7	96.70±1.22	98.64±0.44	99.03±0.90	99.49±0.21	99.61±0.25	99.24±0.19	**99.86±0.04**
8	80.33±2.56	82.65±6.02	74.95±2.42	79.63±3.52	80.92±2.88	79.83±2.83	**85.13±2.35**
9	96.53±1.06	97.71±0.46	95.40±1.39	98.99±0.90	**99.57±0.20**	98.85±0.60	99.50±0.34
10	75.50±4.93	76.15±5.25	87.60±2.48	**95.85±1.77**	83.36±3.41	86.71±3.57	93.01±2.07
11	74.26±6.71	64.33±7.28	86.12±3.89	90.96±4.12	78.33±5.35	87.78±4.63	**94.32±3.87**
12	93.19±3.95	77.53±16.14	89.20±7.10	98.44±1.61	92.96±3.30	97.76±2.17	**99.63±0.44**
13	86.61±4.34	94.69±2.82	87.21±8.00	97.38±1.53	93.26±3.85	97.57±1.62	**98.18±1.58**
14	61.19±7.16	81.75±3.95	83.11±4.57	91.19±3.36	84.31±6.22	89.09±3.10	**92.95±3.20**
15	59.06±3.41	33.24±13.69	60.49±3.67	**63.59±3.91**	53.34±5.33	61.69±4.38	63.48±3.87
16	67.84±8.81	80.17±3.11	96.23±1.29	97.37±2.38	95.03±1.76	93.73±3.49	**98.48±0.62**
OA	83.45±1.17	81.06±0.93	85.25±1.05	89.59±0.66	86.24±0.53	88.34±0.67	**90.99±0.35**
AA	84.11±1.69	83.56±0.95	89.27±1.20	94.10±0.43	89.42±0.76	92.43±0.53	**95.02±0.39**
κ	81.34±1.35	78.80±1.08	83.57±1.17	88.41±0.73	84.63±0.60	87.00±0.75	**89.95±0.39**

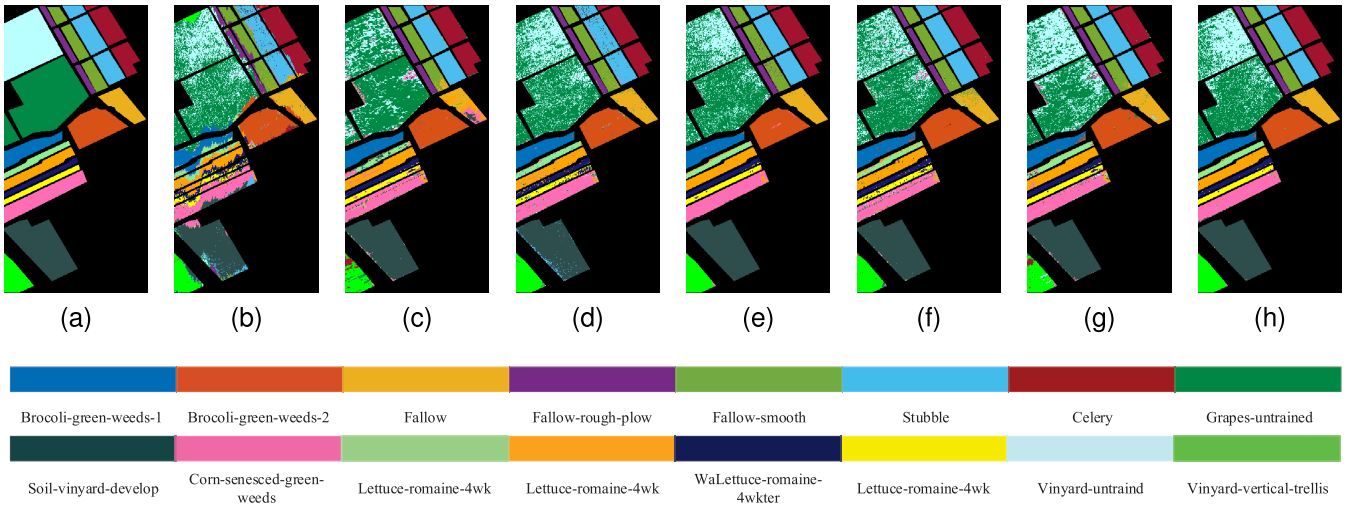


Fig. 8. Classification maps of different methods for the Salinas dataset. (a) Original. (b) LPP. (c) SLGDA. (d) LRE. (e) LPRR. (f) DLRP. (g) RPL. (h) $L_{2,p}$-RER ($p = 0.8$).

accuracy on each category for the Salinas dataset. From Fig. 8, it can be also seen that the map acquired by the proposed method is the closest to the original map. For the Heihe dataset, the highest performance is achieved by $L_{2,p}$-RER with the values of OA = 91.02%, AA = 90.39%, and κ = 87.92%. It gains increases of 8% in OA compared with LPP and over

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TABLE VI
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR THE HEIHE DATASET USING THE SVM CLASSIFIER

Class	LPP	SLGDA	LRE	LPRR	DLRP	RPL	$L_{2,p}$-RER (p=1.4)
1	79.01±4.07	83.19±11.25	83.26±3.99	**89.88±1.68**	88.59±2.77	83.18±4.18	88.69±3.34
2	**99.41±0.45**	87.68±5.01	97.74±0.65	91.54±3.86	91.05±2.91	98.79±0.42	97.13±1.56
3	71.87±5.23	81.95±11.41	79.33±4.03	86.64±4.10	85.70±3.09	77.41±3.57	**88.34±3.43**
4	74.84±5.83	62.41±14.75	78.61±3.16	77.01±8.36	82.75±5.16	78.08±5.17	**87.84±2.95**
5	74.21±6.21	85.89±10.91	84.34±3.66	92.74±3.11	**93.14±2.69**	82.09±3.93	92.45±3.49
6	79.78±5.38	79.94±8.89	84.69±3.42	89.07±3.64	**92.37±1.78**	83.05±5.90	89.78±3.91
7	68.35±8.23	60.18±10.18	70.96±6.14	70.57±10.24	78.47±6.80	71.51±7.28	**85.90±5.78**
8	74.59±5.28	88.36±12.00	82.64±4.91	92.00±2.62	**93.86±3.56**	80.85±4.73	93.00±3.66
OA	82.59±2.21	82.54±7.16	86.05±1.51	88.70±1.12	88.45±1.67	85.78±2.00	**91.02±1.51**
AA	77.76±2.33	78.70±7.77	82.70±1.42	86.18±1.50	88.24±1.54	81.87±1.43	**90.39±1.46**
κ	76.31±2.96	76.81±9.73	81.25±1.89	84.84±1.47	84.54±2.15	80.82±2.55	**87.92±1.95**

TABLE VII
ABLATION ANALYSIS ON THE UNIVERSITY OF PAVIA DATASET

Baseline	✓	✓	✓	✓
ER	✗	✓	✗	✓
$L_{2,1}$-NC	✗	✗	✓	✓
OA	73.87±1.48	80.48±2.01	80.27±1.33	**86.47±0.79**
AA	59.92±2.45	68.20±2.80	75.08±1.69	**82.81±1.42**
κ	63.19±2.28	73.36±3.07	73.55±1.75	**81.83±1.07**

TABLE VIII
ABLATION ANALYSIS ON THE SALINAS DATASET

Baseline	✓	✓	✓	✓
ER	✗	✓	✗	✓
$L_{2,1}$-NC	✗	✗	✓	✓
OA	80.77±1.19	87.13±0.78	85.88±0.97	**90.99±0.35**
AA	83.81±1.46	91.79±0.66	90.76±0.82	**95.02±0.39**
κ	78.59±1.33	85.67±0.86	84.30±1.07	**89.95±0.39**

TABLE IX
ABLATION ANALYSIS ON THE HEIHE DATASET

Baseline	✓	✓	✓	✓
ER	✗	✓	✗	✓
$L_{2,1}$-NC	✗	✗	✓	✓
OA	78.83±4.04	88.17±2.21	86.46±1.85	**91.02±1.51**
AA	72.24±3.27	87.56±1.65	84.11±1.37	**90.39±1.46**
κ	72.20±4.77	84.21±2.77	81.97±2.35	**87.92±1.95**

4% in OA compared to the other state-of-the-art FE approaches (such as SLGDA, LRE, and RPL). Specifically, $L_{2,p}$-RER offers about 2% higher accuracy than LPRR and DLRP, which also integrated the linear regression. This demonstrates that $L_{2,p}$-RER can obtain more robust and discriminative low-dimensional features for classification. In conclusion, the $L_{2,p}$-RER achieves the best classification performance in both quantitative and visual results, which demonstrates that the $L_{2,p}$-norm and the ER can help to extract more robust and discriminative low-dimensional features.

V. MODEL ANALYSIS

A. Ablation Study

In this section, the ablation study is conducted to discuss the effects of the ER term and the $L_{2,1}$-NC on the projection matrix. Thus, the version of model (8) without both ER and $L_{2,1}$-NC is considered as a baseline for comparison. The average OA, AA, and κ values of different datasets under different situations are employed to evaluate the performance of the extracted low-dimensional features. The experimental results on the three datasets are shown in Tables VII–IX. It can be seen that the two components are consistently effective in promoting performance of feature dimensionality reduction. Specifically, for the University of Pavia dataset, compared with the baseline, the vision of baseline+ER yields gains of about 7% in OA, 8% in AA, and 10% in κ. Adding $L_{2,1}$-NC to baseline also achieves a noteworthy improvement of about 7% in OA, 15% in AA, and 10% in κ while comparing to baseline. More importantly, the version of baseline with both ER and $L_{2,1}$-NC (i.e., the proposed $L_{2,p}$-RER model) achieves the best classification performance with OA of 86.47%, AA of 82.81%, and κ of 81.83%. From Tables VIII and IX, the similar conclusion can be drawn for the Salinas and Heihe datasets. Thus, it can be concluded that both ER and $L_{2,1}$-NC play the prominent role for $L_{2,p}$-RER to enhance the robustness of feature dimensionality reduction.

B. Convergence Analysis

In this section, the convergence of the proposed model is tested on the involved three datasets. The associated numerical simulation results are presented in Fig. 10. Since the L, E, and Q are initialized to zero matrix, all curves show an upward trend when the number of iterations is less than 30. Fig. 10(a)–(c) shows the curves of the objective function value of problem (11) on the University of Pavia, Salinas, and Heihe datasets, respectively. Obviously, when the number of iterations is over 30, the value of the objective function gradually decreases with the increase in the number of iterations. After about 100 iterations, all convergence curves tend to be stable. Of course, some additional iterations are needed to continuously self-tune parameters, that is, $L_{2,p}$-RER can efficiently achieve convergence in practice.

C. Complexity Analysis

This subsection gives the computational complexity analysis of Algorithm 1. Suppose that d and n are the dimension of original data and the number of training samples, respectively, and c denotes the number of total classes. The projection P reduces the raw data into a low-dimensional space with m features. For updating M, the main computational complexity comes from the SVT, which needs $O(n^3)$. The progress of computing L will cost $O(n^3 + dn^2 + dmn)$. Then, the computational complexity for solving E, Q, and P is $O(n^3 +$

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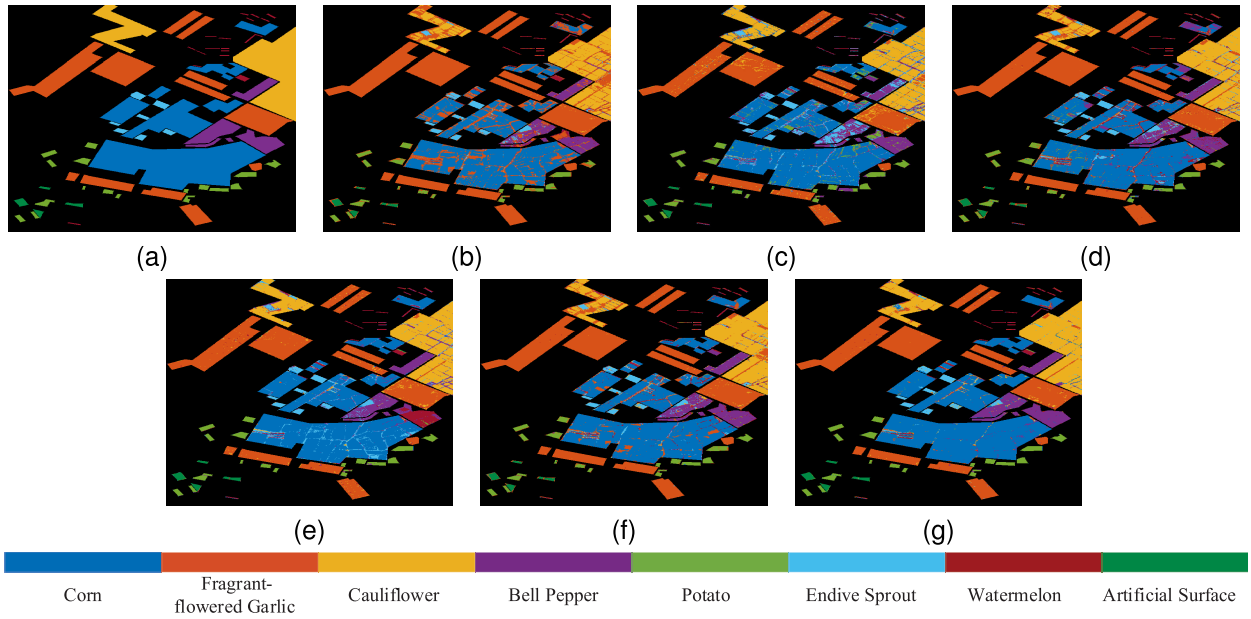


Fig. 9. Classification maps of different methods for the Heihe dataset. (a) Original. (b) LPP. (c) SLGDA. (d) LRE. (e) LPRR. (f) DLRP. (g) RPL. (h) $L_{2,p}$-RER ($p = 1.4$).

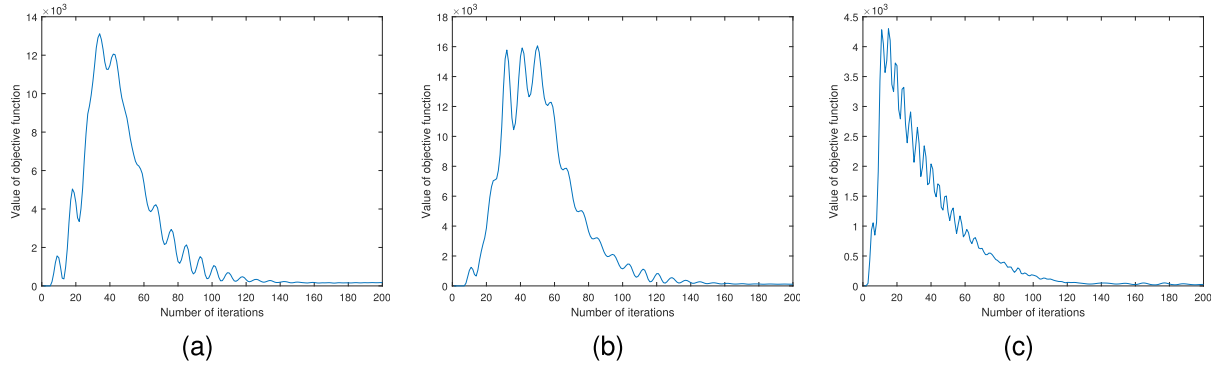


Fig. 10. Convergence of $L_{2,p}$-RER on three databases. (a) University of Pavia dataset. (b) Salinas dataset. (c) Heihe dataset.

TABLE X
COMPUTATIONAL TIME (IN SECONDS) OF DIFFERENT METHODS

	University of Pavia	Salinas	Heihe
LPP	0.67	1.25	0.67
SLGDA	0.85	3.06	1.03
LRE	10.47	22.54	1.89
LPRR	0.67	1.82	0.31
DLRP	4.99	8.50	0.83
RPL	0.43	1.23	0.31
$L_{2,p}$-RER	3.78	7.05	1.02

$mn^2 + dmn$), $O(n^3 + cn^2 + cmd + cnd)$, and $O(n^3 + d^3 + dn^2 + mn^2 + nd^2 + mnd + mnc)$, respectively. For optimizing W in (25), the time complexity consists of two parts: calculating $P^T \mathrm{X} \mathrm{F} \mathrm{Y}^T$ and its SVD decomposition. Computing $P^T \mathrm{X} \mathrm{F} \mathrm{Y}^T$ takes $O(cn^2 + nc^2 + cmn)$ and the complexity of its SVD decomposition is at most $O(c^3)$. Thus, the total complexity of updating W is at most $O(c^3 + cn^2 + nc^2 + cmn)$. Finally, computing D, B, and F cost $O(md)$, $O(mn)$, and $O(cn)$, respectively. In practical applications, n is generally bigger than other variables. Thus, the total computational complexity of $L_{2,p}$-RER is $O(tn^3)$, where t is the number of iterations.

To quantitatively compare the complexity of each method, each method is rerun 50 times and the average running time of the aforementioned methods is reported in Table X. All experiments are carried out using MATLAB on an Intel Core i7-10700 CPU machine with 48 GB of RAM. It can be seen from Table X that the computational complexity of $L_{2,p}$-RER is relatively higher than that of other feature dimensionality reduction methods (i.e., LPRR and RPL) in three HSI datasets. Fortunately, the running time of $L_{2,p}$-RER is still acceptable.

VI. CONCLUSION

This article introduced a new method named $L_{2,p}$-RER for the feature dimensionality reduction of HSIs, which not only integrates projection learning with RPCA together to recover the clean data but also adds an ER term to enhance discriminability. Furthermore, the $L_{2,p}$-norm was adopted to model the noise term and regression residual, which improves the adaptive ability. Finally, the $L_{2,1}$-NC was imposed on the projection matrix to obtain the key features from all of the original ones.

To verify the performance of $L_{2,p}$-RER, extensive comprehensive experiments were conducted on three HSI datasets

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and the corresponding results illustrated that the $L_{2,p}$-RER method achieves superior performance than the comparative ones. Furthermore, the conclusion can be drawn from those experimental results that the proposed method is effective in feature dimensionality reduction of HSIs and has strong adaptability and robustness.

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